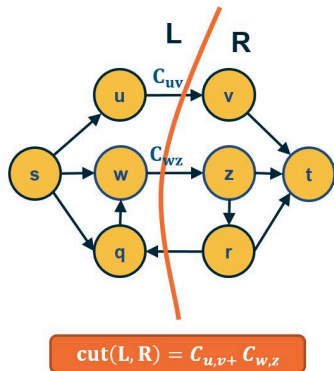


L2: Max-flow=Min-cut

An important result about the min-cut and max-flow problems is that they have the same solution: *the sum of the weights of the min-cut is equal to the max-flow in that network.*



Part A: ANY cut(L,R) such that $s \in L$ and $t \in R$ has capacity $\geq$ ANY flow from s to t.

Thus: $\text{mincut}(s,t) \geq \text{maxflow}(s,t)$

Part B: IF $f^* = \text{maxflow}(s,t)$, the network can be partitioned in two sets of nodes L and R with $s \in L$ and $t \in R$, such that:

- All edges from L to R have flow = capacity
- All edges from R to L have flow = 0.

So, edges from L to R define a cut(s,t) with capacity = maxflow(s,t) and, because of Part A, this cut is mincut(s,t).

Thus: $\text{mincut}(s,t) = \text{maxflow}(s,t)$

